```python
# ============================================================
# BANK LOAN PREDICTION USING NAIVE BAYES
# Google Colab Ready
# ============================================================

# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report


# ------------------------------------------------------------
# 1. Create Bank Loan Dataset
# ------------------------------------------------------------

data = {
    'Age': [25, 35, 45, 28, 52, 40, 30, 48, 23, 55,
            38, 42, 27, 50, 33, 46, 29, 58, 36, 41,
            31, 49, 26, 44, 37, 53, 32, 47, 24, 39],

    'Income': [25000, 45000, 70000, 30000, 90000, 60000, 40000, 80000,
               22000, 95000, 55000, 65000, 28000, 85000, 50000, 75000,
               35000, 100000, 52000, 62000, 42000, 82000, 26000, 68000,
               58000, 92000, 38000, 78000, 24000, 57000],

    'LoanAmount': [100000, 200000, 300000, 120000, 400000, 250000, 180000,
                   350000, 90000, 450000, 220000, 280000, 110000, 380000,
                   190000, 320000, 150000, 500000, 210000, 260000, 170000,
                   360000, 100000, 290000, 240000, 420000, 160000, 340000,
                   95000, 230000],

    'CreditScore': [650, 720, 780, 660, 800, 750, 700, 790, 620, 810,
                    730, 760, 640, 805, 710, 770, 680, 820, 725, 740,
                    695, 785, 630, 765, 735, 795, 675, 775, 615, 745],

    'EmploymentYears': [2, 8, 15, 3, 20, 10, 5, 18, 1, 22,
                        12, 14, 2, 19, 7, 16, 4, 25, 9, 11,
                        6, 17, 1, 13, 10, 21, 5, 18, 2, 12],

    'LoanApproved': [
        'No', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'Yes', 'Yes',
        'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'Yes',
        'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes', 'No', 'Yes',
        'Yes', 'Yes', 'No', 'Yes', 'No', 'Yes'
    ]
}

df = pd.DataFrame(data)

print("Bank Loan Dataset:")
print(df.head())

print("\nDataset Shape:", df.shape)


# ------------------------------------------------------------
# 2. Separate Input and Output
# ------------------------------------------------------------

X = df.drop('LoanApproved', axis=1)
y = df['LoanApproved']


# ------------------------------------------------------------
# 3. Encode Target Variable
# ------------------------------------------------------------

encoder = LabelEncoder()

y = encoder.fit_transform(y)
```

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# No = 0
# Yes = 1

print("\nEncoded Classes:")
print(encoder.classes_)


# ------------------------------------------------------------
# 4. Split Dataset into Training and Testing Data
# ------------------------------------------------------------

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=42,
    stratify=y
)

print("\nTraining Samples:", len(X_train))
print("Testing Samples :", len(X_test))


# ------------------------------------------------------------
# 5. Create Naive Bayes Model
# ------------------------------------------------------------

model = GaussianNB()


# ------------------------------------------------------------
# 6. Train the Model
# ------------------------------------------------------------

model.fit(X_train, y_train)


# ------------------------------------------------------------
# 7. Predict Loan Approval
# ------------------------------------------------------------

y_pred = model.predict(X_test)


# ------------------------------------------------------------
# 8. Calculate Accuracy
# ------------------------------------------------------------

accuracy = accuracy_score(y_test, y_pred)

print("\n===========================================")
print("NAIVE BAYES BANK LOAN PREDICTION")
print("===========================================")

print("\nAccuracy:", accuracy)
print("Accuracy (%):", accuracy * 100)


# ------------------------------------------------------------
# 9. Confusion Matrix
# ------------------------------------------------------------

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)


# ------------------------------------------------------------
# 10. Display Confusion Matrix
# ------------------------------------------------------------

plt.figure(figsize=(6, 5))

sns.heatmap(
    cm,
    annot=True,
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    fmt='d',
    cmap='Blues',
    xticklabels=['Not Approved', 'Approved'],
    yticklabels=['Not Approved', 'Approved']
)

plt.title("Bank Loan Prediction - Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()


# ------------------------------------------------------------
# 11. Classification Report
# ------------------------------------------------------------

print("\nClassification Report:")

print(
    classification_report(
        y_test,
        y_pred,
        target_names=['Not Approved', 'Approved']
    )
)


# ------------------------------------------------------------
# 12. Predict Loan Approval for a New Applicant
# ------------------------------------------------------------

# New applicant details:
# Age = 35
# Income = 60000
# Loan Amount = 200000
# Credit Score = 750
# Employment Years = 8

new_applicant = pd.DataFrame({
    'Age': [35],
    'Income': [60000],
    'LoanAmount': [200000],
    'CreditScore': [750],
    'EmploymentYears': [8]
})

prediction = model.predict(new_applicant)

probability = model.predict_proba(new_applicant)


# ------------------------------------------------------------
# 13. Display New Applicant Prediction
# ------------------------------------------------------------

print("\n==========================================")
print("NEW APPLICANT PREDICTION")
print("==========================================")

if prediction[0] == 1:
    print("Loan Status: APPROVED")
else:
    print("Loan Status: NOT APPROVED")

print("\nPrediction Probabilities:")
print("Not Approved:", probability[0][0])
print("Approved    :", probability[0][1])
```

```
Bank Loan Dataset:
   Age  Income  LoanAmount  CreditScore  EmploymentYears LoanApproved
0   25   25000      100000          650                2           No
1   35   45000      200000          720                8          Yes
2   45   70000      300000          780               15          Yes
3   28   30000      120000          660                3           No
4   52   90000      400000          800               20          Yes

Dataset Shape: (30, 6)

Encoded Classes:
['No' 'Yes']

Training Samples: 21
Testing Samples : 9

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NAIVE BAYES BANK LOAN PREDICTION
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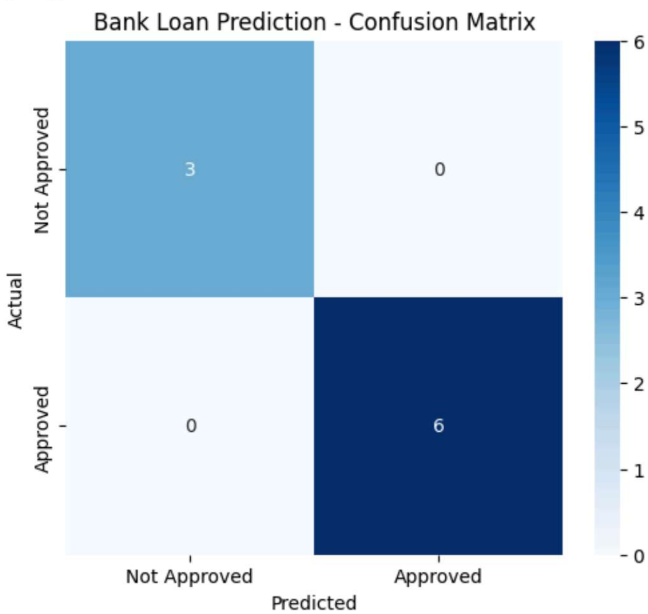
Accuracy: 1.0
Accuracy (%): 100.0

Confusion Matrix:
[[3 0]
 [0 6]]
```




```
Classification Report:
              precision    recall  f1-score   support

Not Approved       1.00      1.00      1.00         3
    Approved       1.00      1.00      1.00         6

    accuracy                           1.00         9
   macro avg       1.00      1.00      1.00         9
weighted avg       1.00      1.00      1.00         9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NEW APPLICANT PREDICTION
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Loan Status: APPROVED

Prediction Probabilities:
Not Approved: 5.571098301969077e-07
Approved    : 0.9999994428901704
```